

ESG-RAG-Index

A Retrieval-Augmented Generation (RAG) framework for constructing firm-level ESG Disclosure Index from Annual Reports, Sustainability Reports, and Integrated Reports using Large Language Models (LLMs).

Overview

This project develops an AI-based ESG scoring system that extracts, evaluates, and quantifies Environmental (E), Social (S), and Governance (G) disclosures from corporate reports.

Unlike traditional dictionary-based ESG measures that rely on keyword counts, this framework evaluates disclosure quality, quantitative evidence, targets, and performance using Retrieval-Augmented Generation (RAG) and Large Language Models.

Pipeline:

```
Annual Reports
  ↓
PDF Parsing / OCR
  ↓
Text Chunking
  ↓
Embedding (BGE-M3 / E5)
  ↓
Vector Database (FAISS)
  ↓
RAG Retrieval
  ↓
LLM Evaluation
  ↓
Indicator Scores
  ↓
E / S / G Subindices
  ↓
ESG Disclosure Index
```

Research Objective

Construct a firm-year ESG Disclosure Index that:

- Captures disclosure quality rather than keyword frequency
- Reduces measurement error from dictionary methods
- Provides transparent evidence for each score

- Supports large-scale panel datasets

Example:

Firm	Year	ESG
Firm A	2020	0.67
Firm A	2021	0.72
Firm B	2020	0.43

Directory Structure

```
esg-rag-index/  
├── data/  
│   ├── raw_reports/  
│   ├── parsed_reports/  
│   ├── chunks/  
│   └── embeddings/  
├── indicators/  
│   ├── environment.csv  
│   ├── social.csv  
│   └── governance.csv  
├── vectorstore/  
│   └── faiss/  
├── prompts/  
│   ├── scoring_prompt.txt  
│   └── extraction_prompt.txt  
├── outputs/  
│   ├── indicator_scores/  
│   ├── ESG_scores/  
│   └── evidence/  
└── src/  
    ├── parser.py  
    ├── chunker.py  
    ├── embedder.py  
    ├── retriever.py  
    ├── scorer.py  
    ├── aggregator.py  
    └── validation.py
```

|
└─ README.md

Step 1. Data Collection

Input documents:

- Annual Reports
- Sustainability Reports
- ESG Reports
- Integrated Reports

Supported formats:

PDF
PDF (Scanned)
HTML
TXT
DOCX

Expected file naming:

FPT_2024.pdf
VIC_2024.pdf
HPG_2024.pdf

Step 2. OCR and Parsing

Convert reports into machine-readable text.

Recommended tools:

- PyMuPDF
- pdfplumber
- Tesseract OCR
- PaddleOCR

Output:

```
{  
  "page": 37,  
  "section": "Environmental Management",  
}
```

```
{  
  "text": "The company reduced Scope 1 emissions..."  
}
```

Step 3. Chunking

Split reports into semantic chunks.

Recommended settings:

```
Chunk Size: 500-800 tokens  
Overlap: 100 tokens
```

Example:

```
Chunk_001  
Chairman's Message  
  
Chunk_002  
Environmental Policies  
  
Chunk_003  
Carbon Emissions
```

Output:

```
{  
  "chunk_id": 3,  
  "company": "FPT",  
  "year": 2024,  
  "text": "The company reported..."  
}
```

Step 4. Embedding Generation

Transform chunks into vector representations.

Recommended models:

Option 1

BGE-M3

Advantages:

- Multilingual
- Strong retrieval performance
- Vietnamese support

Option 2

multilingual-E5-large

Advantages:

- Cross-language retrieval
- Strong semantic matching

Output:

```
embedding.shape  
  
(1024,)
```

Store:

```
embeddings.npy
```

Step 5. Vector Database

Index embeddings.

Recommended:

- FAISS
- ChromaDB
- Qdrant

Example:

```
index.add(embeddings)
```

Stored data:

```
chunk_id  
company
```

Step 6. ESG Indicator Framework

Indicators are derived from:

- GRI Standards
- SASB Standards
- TCFD Framework
- ISSB Standards

Example:

Environmental

ID	Indicator
E1	Carbon emissions
E2	Energy consumption
E3	Renewable energy
E4	Waste management
E5	Water usage

Social

ID	Indicator
S1	Employee training
S2	Diversity
S3	Workplace safety
S4	Human rights
S5	Community engagement

Governance

ID	Indicator
G1	Board independence
G2	Anti-corruption
G3	Audit committee

ID	Indicator
G4	Executive compensation
G5	Shareholder rights

Recommended total:

40-60 indicators

Step 7. Retrieval-Augmented Generation

For each ESG indicator:

Question:

Does the company disclose carbon emissions?

Retriever:

top_k = 5

Output:

Chunk 37
 Chunk 41
 Chunk 85
 Chunk 92
 Chunk 114

These chunks become evidence for LLM evaluation.

Step 8. LLM Scoring

Recommended models:

- GPT-4o
- Claude Opus
- Claude Sonnet
- Llama 3 70B

Prompt:

You are an ESG analyst.

Question:

Does the company disclose carbon emissions?

Evidence:

[Retrieved Chunks]

Score:

0 = No disclosure

1 = General statement

2 = Quantitative disclosure

3 = Quantitative disclosure with targets and progress

Return JSON only.

Output:

```
{
  "indicator": "E1",
  "score": 3,
  "confidence": 0.91,
  "evidence": "Scope 1 and Scope 2 emissions disclosed."
}
```

Step 9. Indicator Scoring

Indicator scale:

Score	Meaning
0	No disclosure
1	Narrative disclosure
2	Quantitative disclosure
3	Quantitative + target + performance

Maximum score:

$$60 \text{ indicators} \times 3 = 180$$

Step 10. Construct E, S, G Subindices

Environmental:

$$E = \Sigma(\text{E indicators}) / \text{Max}(E)$$

Social:

$$S = \Sigma(\text{S indicators}) / \text{Max}(S)$$

Governance:

$$G = \Sigma(\text{G indicators}) / \text{Max}(G)$$

Range:

$$0 - 1$$

Step 11. ESG Disclosure Index

Equal weighting:

$$\text{ESG} = (E + S + G) / 3$$

Alternative:

$$\text{ESG} = w_{EE} + w_{SS} + w_{GG}$$

where:

$$wE + wS + wG = 1$$

Possible weighting methods:

- Equal weighting
 - PCA weighting
 - Entropy weighting
-

Step 12. Validation

Recommended validation procedures:

Human Validation

Sample:

10%

of firm-year observations.

Independent ESG experts score reports manually.

Compute:

Cohen's Kappa
Krippendorff Alpha

External Validation

Compare with:

- Refinitiv ESG
- MSCI ESG
- Sustainalytics

Expected:

Positive correlation

Final Output

Firm-Year ESG Dataset

```
firm,year,E,S,G,ESG
FPT,2024,0.71,0.68,0.82,0.74
VIC,2024,0.63,0.55,0.70,0.63
HPG,2024,0.58,0.61,0.66,0.62
```

Advantages over Traditional ESG Measures

Method	Context Understanding	Quality Evaluation
Keyword Count	No	No
Dictionary-Based	Limited	No
TF-IDF	Limited	No
FinBERT	Moderate	Partial
RAG + LLM	Strong	Strong

Recommended Research Workflow

```
Annual Reports
  ↓
OCR
  ↓
Chunking
  ↓
BGE-M3 Embeddings
  ↓
FAISS
  ↓
RAG Retrieval
  ↓
GPT-4o Scoring
  ↓
E, S, G Scores
  ↓
ESG Disclosure Index
  ↓
Panel Data Analysis
  ↓
Publication
```